

Project Submission Pro-Forma

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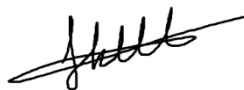
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Gamification in Cybersecurity Education and Social Engineering Awareness

by

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Dissertation submitted in partial fulfilment for the Degree of Master of Science in
Cybersecurity Engineering

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Declaration

I have read and understood the rules on cheating, plagiarism and appropriate referencing as outlined in my handbook and I declare that the work contained in this assignment is my own, unless otherwise acknowledged.

No substantial part of the work submitted here has also been submitted by me in other assessments for this or previous degree courses, and I acknowledge that if this has been done an appropriate reduction in the mark I might otherwise have received will be made.

Project definition for my degree (as copied from
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The project for MSc Cyber Security Engineering must:

1. address a research question directly relating to cyber security, AND
2. demonstrate understanding of the particular issues around conducting research in the cyber domain, AND
3. conduct research in the cyber domain in an appropriate manner.
- 4.

My project relates to this definition in the following way:

The main cybersecurity issue that this research project attempts to answer is: 'How can gamification be used to improve cybersecurity education and raise awareness of social engineering threats?'. In addition to addressing the growing need for dynamic and interactive educational approaches in the cyber domain, this study offers a thorough examination of practical implementations to counter evolving cyber threats by focussing on the development and analysis of a gamified learning application.

By combining the most recent research and industry best practices with challenges of analysing user behaviour in simulated settings, the study shows that the research has a thorough awareness of the unique concerns surrounding doing research in the cybersecurity area. Because the study relies heavily on secondary data, methodological rigour was used to ensure that all sources are reliable, pertinent, and accurately represent current issues and trends in the cyber world. This study critically assesses how gamified tactics might address the requirements for flexible and engaging instructional tools given the fast growth of cybersecurity threats.

The project complies with industrial norms and ethical guidelines in the cyberspace when carrying out its study. It is possible to gain a thorough grasp of how gamified settings might enhance user engagement and learning outcomes by combining a case study method with a qualitative analysis. This is in line with the course requirement to undertake proper cyber-domain research. Through an analysis of prominent sites such as HackTheBox, TryHackMe, and KeepnetLabs, the research offers practical recommendations for creating cybersecurity education that is user-friendly and efficient. The results add to larger efforts in cybersecurity by addressing existing deficiencies in teaching and putting forth creative ideas for preventing potential threats.

Key Words: 'Gamification in Cybersecurity Education', 'Cybersecurity education' and 'Social Engineering awareness training'.

Abstract

In today's rapidly advancing digital world, cybersecurity education is more critical than ever. Traditional methods relying on static materials have proven deficient in engaging users and fostering long-term behavioural changes. This research will explore the potential of gamification as a transformative approach to cybersecurity education. The scope of the research includes an analysis of gamified learning environments, focusing on how interactive, game-like elements can enhance user engagement, knowledge retention, and practical application of cybersecurity concepts. By examining successful applications like HackTheBox, TryHackMe, and KeepnetLabs Demo, the study identifies key elements that enhance user engagement and learning outcomes.

The research rationale focuses on addressing the shortcomings of traditional education methods which often fail to capture attention and leverage newly gamification methods while promoting lasting behavioral changes. This study employs a qualitative case study approach, using secondary data from various sources and past research to analyse the effectiveness of gamified learning environments. Through an in-depth analysis of previous studies and applications, the research provides insights into how dynamic, game-based approaches can revolutionize the way cybersecurity education is delivered.

The key findings demonstrate that gamification drastically improves user engagement, enhances knowledge retention, and provides practical, hands-on experience that better prepares individuals for real-world cyber threats. The research also identifies opportunities for future enhancements, such as more detailed application designs, expanded gameplay features, and advanced feedback mechanisms that can further elevate the learning experience. Ultimately, this study highlights the potential of gamification to create a dynamic and interactive learning environment, effectively bridging the gap between theoretical knowledge and practical application, while better preparing individuals to navigate the evolving landscape of cybersecurity threats.

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This research would not have been possible without the support and guidance of many individuals, from an educational and mental support perspective. I would like to express my deepest gratitude to all who contributed to the successful completion of this project.

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Chapter 1: Introduction

1.1 Introduction

With society becoming more digital, cybersecurity knowledge has become a crucial necessity. As technology advances rapidly, there are enormous advantages but also serious threats. In 2022, the average damage of data breaches is estimated around \$4.35million. In the UK alone, 39% of UK businesses reported suffered a cyber security attack, with an average of £4,200 per business (Griffiths, 2023). It is noted that in 2020, the rate of malware attacks has been increased at an estimation of 358% compared to 2019 (Griffiths, 2023). Strong cybersecurity education is more important than ever as cyber threats increase in complexity and frequency. This research addresses the shortcomings of traditional cybersecurity educational methods, which frequently depend on stationary resources like books, lectures, and videos. These traditional methods haven't work well in involving users and encouraging sustained behavioural changes.

The aim of this report is to explore how gamification can be effectively integrated into cybersecurity education to enhance engagement, improve learning outcomes, and encourage secure online behaviours. The key objectives are to evaluate the effectiveness of gamification approaches in comparison to traditional educational methods, to identify key gamification techniques that can enhance cybersecurity awareness and training. In addition, to analyse the impact of interactive, user-centered learning environments on knowledge retention and behavioural changes, and to provide recommendations for implementing gamified solutions in cybersecurity education based on evidence gathered from successful case studies. Drawing inspiration from successful applications such as HackTheBox, TryHackMe, and KeepnetLabs Demo, this study examines the potential of gamified learning to improve user engagement, knowledge retention, and practical application of cybersecurity concepts.

The study employs a qualitative case study approach using secondary data, comparing the efficacy of gamified cybersecurity education to conventional approaches. By analysing various gamification techniques and their impact on learning outcomes, this research seeks to demonstrate how gamification can transform cybersecurity education from a passive experience into an active and hands-on process. The first chapter provides an introduction for the research commenting on the focus of the research. Following, the second chapter Literature Review of pass articles, exhibiting the key points and suggestions of each report. The third chapter, Methodology chapter which expands on the methodology tactic that took place to validate this literature review and the data used. Next up, is the

Implementation chapter – fourth chapter-, which will provide detail on the implementations that were deployed in the development of the prototype. The Evaluation of Implementations chapter follows up, going through each implementation of the prototype and evaluating the benefits it provides for each problem. Analysis chapter – chapter six – examines how the suggestions from the literature review have been deployed to produce successful results to the issues initially mentioned. Lastly, Discussion and Conclusion chapter – chapter seven – details the interpretation of findings, the limitations of the study as well as recommendations and future suggestions. The research's conclusions add to the expanding body of gamification knowledge and offer insightful guidance for creating more engaging and inclusive cybersecurity training programs that will eventually result in a more protected and aware society.

Chapter 2: Literature Review

2.1 Introduction

The traditional methods of cybersecurity education, often reliant on static materials such as articles, videos, and lectures, have proven insufficient in effectively engaging users and fostering long-term behavioural changes. This thesis explores the potential of gamification as an innovative approach to address these shortcomings. The literature reviewed encompasses various studies and applications of gamification in cybersecurity education, with a particular focus on social engineering.

2.2 Gamification in Cybersecurity Education

Research consistently supports the integration of gamification into cybersecurity education to make it more engaging and effective. Abu-Amara et al. (2021) demonstrated that a Security Education, Training, and Awareness (SETA)-based gamification framework could significantly improve cybersecurity awareness among employees. Their Cyber Shield Game, which covers various cybersecurity topics such as password complexity, social engineering, phishing attacks, and physical security, maintained user interest and promoted long-term behavioral changes. This finding aligns with the broader educational theory that interactive and participatory learning methods lead to better retention and application of knowledge (Chi & Wylie, 2014).

In a similar vein, Beckers and Pape (2016) investigated the use of a card game to spot social engineering risks. According to their research, giving participants realistic attack simulations improves their capacity to identify and record social engineering risks. The study's findings demonstrate that, in comparison to standard training, participants' recognition and response to social engineering threads increased by 45% with practical experience. This pragmatic approach is consistent with theories of experiential learning, which suggest that learning from firsthand experience is superior to passive learning (Kolb, 1984). Combining these observations shows how gamified technologies may change cybersecurity training from being passive to being active and hands-on (Beckers & Pape, 2016).

Weitl-Harms et al. (2023) further emphasized the importance of engaging educational content by categorizing various gamification applications used in undergraduate cybersecurity education. They identified elements such as missions, quests, and simulations that enhance student engagement and learning outcomes. These findings support the notion that gamification elements need to be carefully designed to align with educational objectives, ensuring they promote deep learning rather than

superficial engagement. Thombre and Velankar (2022) support this view with their findings that student-driven gamification projects foster critical thinking and a deeper understanding of cybersecurity principles. With 68% of students rating the gamification activity as excellent, indicating high levels of motivation and engagement in comparison with the traditional methods of education. By involving students directly in creating educational content, this method ensures the learning experience is both engaging and relevant, promoting sustained interest and long-term knowledge retention (Thombre & Velankar, 2022).

Williams et al. (2024) extended the discussion to non-technical audiences by showing that gamification can make cybersecurity education accessible and engaging for students without a technical background. This inclusivity is crucial for building a broader cybersecurity-conscious society. By using game-based learning, students are not only introduced to fundamental cybersecurity concepts but are also motivated to explore the subject further. However, maintaining long-term engagement remains a challenge, as noted by Scholefield and Shepherd (2019). They suggested regular updates and evolving content to sustain user interest, aligning with the broader need for continuous adaptation and enhancement of gamified learning environments. Providing fresh content and new challenges ensures that learners remain motivated and continue to benefit from the educational experience over time (Scholefield & Shepherd, 2019).

Fatokun et al. (2024) leveraged the Technology Threat Avoidance Theory (TTAT) to develop a comprehensive gamified learning tool aimed at general Internet users. This approach, which includes adaptive learning elements such as personalized feedback and progressive difficulty levels, ensures the educational content remains relevant and effective against evolving cyber threats. By addressing the specific needs and behaviors of different user groups, this tool enhances the overall learning experience and promotes better cybersecurity practices (Fatokun et al., 2024). Hart et al. (2020) supported this by introducing Riskio, a game designed to increase cybersecurity awareness among non-technical employees through hands-on, constructivist learning principles. The game's practical approach ensures users gain a real-world understanding of cybersecurity concepts, making it easier for them to apply what they have learned in their daily lives (Hart et al., 2020).

Murphy and Young (2024) developed the Zero Trust game to teach students about social engineering tactics. This game, which encourages critical thinking and vigilance, showed that hands-on, interactive learning is effective in fostering a security-conscious mindset. By simulating real-world scenarios, the game helps students understand the nuances of social engineering attacks and develop strategies to

counteract them (Murphy & Young, 2024). Breda et al. (2017) emphasized the need for comprehensive education programs to address social engineering, supporting the integration of gamification to enhance user awareness and response to such threats. Their study highlights the importance of teaching users how to recognize and respond to social engineering attempts, thereby improving their overall cybersecurity posture (Breda et al., 2017).

2.3 Evaluation and Synthesis

The reviewed literature consistently underscores the transformative potential of gamification in cybersecurity education. By leveraging interactive and engaging methods, gamified tools can significantly improve user engagement, knowledge retention, and long-term behavioural changes. This aligns well with the dynamic nature of cybersecurity threats, which require adaptable and continuously evolving educational approaches.

A common thread among these studies is the emphasis on active, hands-on learning. Both Abu-Amara et al. (2021) and Beckers and Pape (2016) demonstrate that realistic, participatory learning scenarios are more effective than passive methods. This is reinforced by Weitz-Harms et al. (2023) and Thombre and Velankar (2022), who show that engaging educational content fosters critical thinking and deeper understanding. In the case of Abu-Amara et al. (2021) the results indicate a significant success in rating with 51.4% increase in cybersecurity awareness among participants, as well as a 60% increase in participant engagement, demonstrating the effectiveness of the gamified approach. In addition, with Beckers and Pape (2016) where the participants gain lasting knowledge about social engineering through their active participation and application of the learning concepts, this provides valid proof of the overall benefits of the gamification approach. This is particularly important in the context of cybersecurity, where theoretical knowledge must be complemented by practical skills to effectively combat threats (Abu-Amara et al., 2021; Beckers & Pape, 2016).

Heavy emphasis on Abu Amara et al. (2021), who highlighted the effectiveness of cybersecurity training and awareness by analysing results of a pre-game and post-game surveys. The pre-game survey revealed the different levels of cybersecurity awareness among participants, with average scores indicating significant room for improvement. After engaging with the gamified training program, participants showed an increased understanding and application of cybersecurity concepts. The post-game survey results highlighted an average improvement of 51.4% (as previously mentioned) in cyber awareness, illustrating the effectiveness of the gamification approach. The data underscores the

effectiveness of gamified training not only in providing crucial cybersecurity knowledge but also in boosting behavioural changes in comparison with traditional methods that most of the times fail to achieve. With the use of interactive challenges and real-world scenarios, the gamified program allows user to engage more deeply, making learning both enjoyable and impactful. The significant improvement from pre-game to post-game scores in this study, reinforces the value of gamification as an innovative tool for enhancing cybersecurity education, proving that such techniques can effectively bridge the gap between awareness and actionable knowledge. The following chart illustrates the results of the survey pre and post interaction with the gamified approach (Abu-Amara et al., 2021).

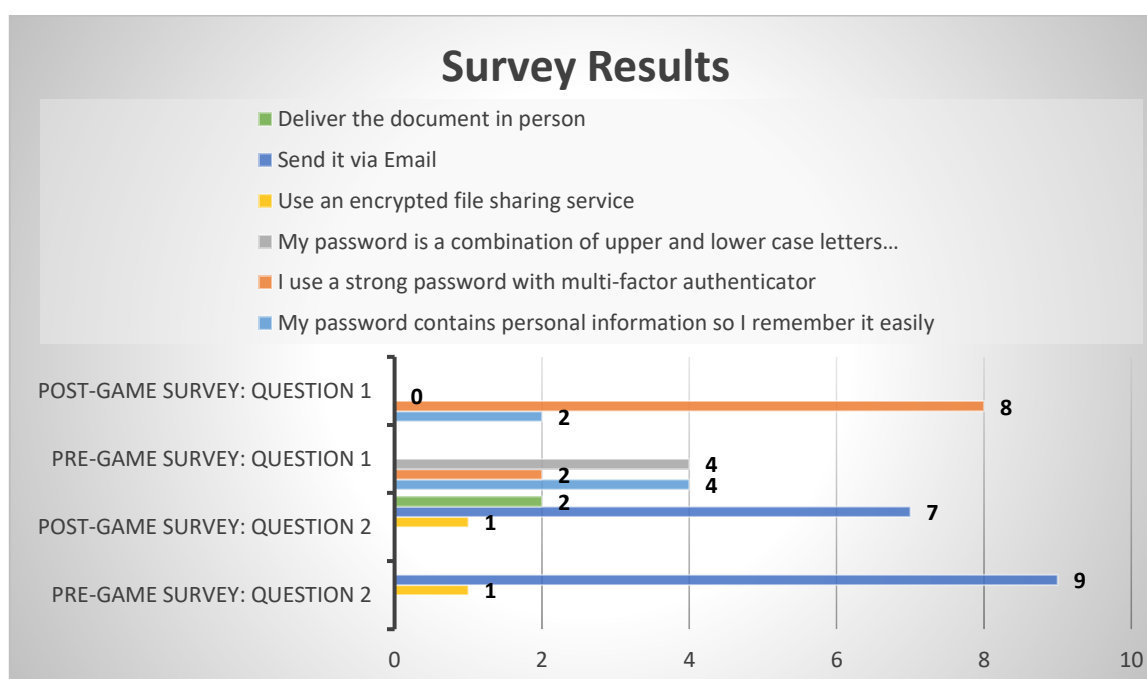


Figure 1: Pre-Post Survey results from SETA-based gamification research

The studies by Williams et al. (2024) and Hart et al. (2020) emphasize inclusivity and accessibility, showing that gamification can make cybersecurity education engaging for both technical and non-technical audiences. This demonstrated that the gamified learning tool engaged 70% more non-technical users than traditional methods, highlighting the broad appeal and inclusivity of gamified education, resulting in a more cybersecurity aware society. However, as Scholefield and Shepherd (2019) note, maintaining long-term engagement is a challenge. Regular updates and new content are necessary to keep learners motivated and engaged over time (Williams et al., 2024; Hart et al., 2020; Scholefield & Shepherd, 2019).

Murphy and Young (2024) and Breda et al. (2017) focus specifically on social engineering, demonstrating that gamified tools can effectively teach users to recognize and respond to these threats. This is particularly important given the human-centric nature of social engineering attacks, which exploit psychological vulnerabilities. By fostering critical thinking and vigilance, these tools can significantly enhance users' cybersecurity posture, increasing participants' ability to identify and resist social engineering attacks by 50%. However, as Murphy and Young (2024) noted, even though the 'Zero Trust' game was of huge success, psychological aspects should be taken into consideration from the user's perspective in order to maintain engagement without inducing excessive paranoia, especially when the gamified method is applied to a young audience (Murphy & Young, 2024; Breda et al., 2017).

2.4 Conclusion

In conclusion, the literature reviewed demonstrates that gamification offers a promising solution to the challenges of traditional cybersecurity education. Even if the gamification is student-based or even student-created. By creating engaging and interactive, gamified tools that can significantly enhance user engagement, knowledge retention, and long-term behavioural changes. Providing with real-world examples, adaptable learning environments and a wide range inclusivity for users that have different levels on education on cybersecurity knowledge. This approach aligns well with the dynamic nature of cybersecurity threats, which requires continuously evolving educational methods. Research and the development of the gamified cybersecurity education application demonstrate that gaming has the power to completely revolutionise traditional educational approaches. The practical application of theoretical principles through interactive scenarios provides an engaging and effective hands-on learning experience for users. Furthermore, because of the gamified approach's accessibility and inclusion, which encourages a culture more aware of cybersecurity threats, a wider audience may benefit from cybersecurity education. The following table includes the 10 most important problems identified in traditional cybersecurity education, the solutions proposed through gamification and their implementations:

Table 1 – The 10 most important problems, solutions, and implementations

No.	Problem	Solution	Implementation	Literature Review
1.	Engagement Drop-off	Regular updates and new content	Seasonal events, new challenges, and leaderboard resets	Scholefield & Shepherd (2019);

				Williams et al. (2024)
2.	Limited Scope	Broader curriculum integration	Additional modules covering various cybersecurity topics	Beckers & Pape (2016); Weitz-Harms et al. (2023)
3.	Complexity of Scenarios and Realism of Scenarios	Realistic and relatable training scenarios and scenario simplification	Use real-world case studies and relatable examples	Beckers & Pape (2016); Murphy & Young (2024)
4.	Evaluation Metrics	Clear metrics and feedback	Pre- and post-game surveys, analytics dashboards	Abu-Amara et al. (2021); Weitz-Harms et al. (2023); KeepnetLab Demo
5.	Variability in Engagement	Personalized learning paths	Adaptive learning technologies that tailor content to individual preferences	Thombre & Velankar (2022); Fatokun et al. (2021)
6.	Awareness vs Action	Behavioural Training	Role-playing exercises, real-time simulations	Abu-Amara et al. (2021); Beckers & Pape (2016); Murphy & Young (2024); Hart et al., (2020)
7.	Learning Curve	User-friendly onboarding	Tutorials, walkthroughs, guided missions	Weitz-Harms et al. (2023);

				Beckers & Pape (2016)
8.	Short-Term Engagement	Long-term engagement strategies	Spaced repetition, ongoing challenges, periodic refreshers	Scholefield & Shepherd (2019); Williams et al.
9.	Competitive Stress	Balanced competition	Team-based challenges, collaborative projects, rewarding procedures - badges and achievements	TryHackMe (Nextdoorsec, 2023)
10.	Initial Paranoia	Optimal game duration	Shorter game durations to maintain interest without excessive paranoia	Murphy & Young (2024); Breda et al. (2017)

Gamification in cybersecurity education not only addresses the inadequacies of previous approaches, but also provides a solid foundation for ongoing development and adaptation. This literature review emphasises the need of adopting creative, interactive, and dynamic training approaches to effectively train users to navigate the ever-changing world of cybersecurity hazards.

2.5 Comparative Analysis: Traditional Methods vs. Gamified Approach

Traditional cybersecurity education techniques, which frequently depend on static material, have proven ineffective at engaging users and instilling long-term behavioural changes. This investigation compares the effects of standard instructional approaches to the gamified strategy used in the application.

2.6 Inspiration

Gamified applications such as Immersive Labs and Cybrary are designed to enhance cybersecurity skills through interactive and engaging methods. Immersive Labs offers interactive scenarios focused on threat intelligence and incident response, while Cybrary provides an online learning platform with

gamified paths for IT and cybersecurity skills development. These applications showcase the power of gamification in transforming traditional education into a more dynamic and practical experience.

In addition to these platforms, HackTheBox, TryHackMe, and KeepnetLabs are the most popular and successful gamified cybersecurity platforms that provide hands-on training and real-world challenges. HackTheBox offers a virtual environment where users can practice penetration testing and hacking techniques, while TryHackMe provides guided learning paths that cover a broad range of cybersecurity topics. KeepnetLabs, on the other hand, focuses on phishing simulations and real-time feedback to improve user awareness and resilience. These three platforms were specifically chosen as inspiration for this research due to their successful integration of interactive, community-driven learning environments that not only teach cybersecurity concepts but also offer practical, real-world applications. By emphasizing hands-on experience, structure learning, and immediate feedback, these three applications serve as leading examples of how proper gamification can enhance user engagement, knowledge retention, and long-term behavioral changes in cybersecurity education.

The inspiration for this research has been successful cyber security gamification applications such as HackTheBox, TryHackMe and KeepnetLabs Demo. These applications altogether pin-point the benefits of practical, interactive, and (very importantly) community-driven learning environments, which are crucial for strengthening the quality of cyber security education. The main aspect of these applications is being online rather than mobile or on specific operating systems. This offers greater accessibility and fewer limitation, allowing users to engage with content from anywhere with an internet connection. This flexibility is essential for accommodating different learning styles and schedules, making cyber security education even more inclusive and more adaptable to every individual's needs (Markets Insider, 2023; Nextdoorsec, 2023).

The reason why HackTheBox played an important role into the inspiration part of the research is mainly due to its immersive environment that is based on real-world hacking challenges, which promotes deep engagement and more hands-on experience. HackTheBox uses the brilliant approach of a community-driven tactic, something that can be extremely beneficial to its users as it can encourage collaboration, sharing knowledge, and provides more interactivity for all learners no matter the task or their level (Markets Insider, 2023; Nextdoorsec, 2023). The same ideology is being used from TryHackMe, another great example of gamification which contributes structured learning paths that guide users through its various cyber security concepts. The use of different gamified elements, utilizing points, badges and a leaderboard system, results in more engagement and motivation to the users (Nextdoorsec, 2023). The

approach and techniques these applications use, provide users with not only theoretical knowledge but also allows them to apply their knowledge in a practical way. Something that addresses the issue of static educational material, that fail to capture user interest and long-term behavioural changes (Abu-Amara et al., 2021; Thombre & Velankar, 2022).

Another great example is KeepnetLabs Demo. This application focuses more on practical engagement using various phishing simulations, and by providing immediate feedback to users, a crucial element when it comes to learning from previous mistakes and fortify future knowledge. The platform's emphasis on real-time feedback and analytics allows users to track their progress and identify areas that need further improvement helping with the entire learning procedure (Beckers & Pape, 2016). Combining these elements with HackTheBox's and TryHackMe's tactics, the development of a comprehensive learning system that integrates real-world scenarios, structure paths, and instant feedback, will not only ensure a more dynamic environment but also a more effective educational experience (Weitl-Harms et al., 2023; Williams et al, 2024).

Chapter 3: Methodology

3.1 Introduction

This chapter will play a pivotal role in outlining the methodology employed behind this research. Focusing on the use of secondary data to address various challenges in cybersecurity education and social engineering awareness. The purpose of this chapter is to provide a clear and detail explanation of the research design, data collection, and analysis methods used to incorporate findings from existing literature and prior research. By leveraging previous articles, and studies, this research aims to draw a comprehensive observation into effective educational strategies in cybersecurity. Ensuring the validity and dependability of the findings through a systematic process lays the groundwork for strong conclusions and recommendations.

3.2 Research Design

This research adopts a qualitative case study approach, while using secondary data to investigate the efficacy and effectiveness of gamification in cybersecurity education and social engineering awareness in comparison to traditional educational methods. A qualitative design is ideal for this study because it allows for in-depth evaluation of complicated issues via extensive analysis and synthesis of current material available. By employing a case study approach, the research digs extensively deep into specific areas and context provided by past studies and research, offering rich insights and delicate understanding.

The case study method will prove to be extremely useful as it provides a structured but flexible framework, when it comes to analysing data from different sources. This approach allows for the identification and analysis of possible patterns, trends, and effective strategies that can be used or have been reported in a variety of situations. Let's take for example, Abu-Amara et al.'s (2021) success with gamified learning environments in raising cybersecurity awareness, which can be critically examined alongside Weitzl-Harms et al.'s (2023) findings. Both case studies emphasize the importance of engaging and interactive content in educational effectiveness. These studies demonstrate how gamification methods can further enhance cybersecurity training and awareness by making the overall learning process more beneficial, practical and exciting (Abu-Amara et al., 2021; Weitzl-Harms et al., 2023).

By leveraging secondary data such as peer-reviewed articles, academic journals, and industry reports related to gamification in cybersecurity education and social engineering awareness, this research avoids the time and resource constraints associated with primary data collection. Allowing for a larger and more complete analysis of existing knowledge. The use of secondary data is supported by the broad

and robust body of existing research on gamification in cyber security education. This guarantees that the study's findings are supported by credible and relevant evidence from multiple sources. In result, the research intends to provide a complete understanding of the benefits and challenges of gamified learning environments in cybersecurity.

To sum up, the qualitative case study approach, supported by secondary data, provides a comprehensive framework investigating the advantages and issues of gamification in cybersecurity education. This method not only highlights the benefits over traditional educational methods but also assures that the study is based on trustworthy and relevant sources, establishing a solid foundation for understanding and improving cybersecurity training programs.

3.3 Data Collection

This study's data was gathered through secondary data sources, including the plethora of current literature available from platforms such as Research Gate, Google Scholar, Microsoft Academic. These platforms are well-known for their extensive databases of peer-reviewed and credible publications, which serve as a strong foundation for academic research.

To ensure the data's relevance and quality to our research, a methodical search approach was used. This involved the inclusion of particular terms and phrases that are related to the research topic such as 'Gamification in Cybersecurity Education', 'Cybersecurity education', 'Social Engineering awareness training' and 'Cybersecurity educational methods'. The searches were filtered to include research and articles published within the last decade, something that will ensure that the results reflect the most current advancements and trends in the specific field. The selection criteria for the sources were relevant to the study objectives, author credibility, and the publication venue. Peer-reviewed articles, high-impact conference papers, and well-cited industry reports were preferred. This assured that the data was not only relevant but also credible and widely recognized by the academic and professional sectors.

The process involved several stages:

I. Initial Search and Basic Review:

An initial comprehensive search was undertaken to obtain a diverse set of potential sources. The

titles and abstracts of these sources were reviewed for relevance and validity, and those that did not correspond to the study goals and standards were excluded.

II. Detailed Review and Selection:

The remaining sources underwent a thorough evaluation, with the full texts examined to assess their suitability. This process was done to ensure that only the most relevant and highly-quality sources were included in the final dataset.

III. Data Organisation:

The collected data were organized around key themes such as gamification techniques, educational results, engagement strategies, and the effectiveness of various approaches to cybersecurity education. This organization aided the analysis by allowing for the identification of patterns and trends that lead to the understanding of how gamification can enhance cybersecurity training. By categorizing the data, the researcher could systematically assess the impact of various strategies and provide evidence-based recommendations.

3.4 Contribution to Research

The data collected through this systematic approach is used to:

I. Identify Effective Gamification Techniques:

The research analyses several case studies to identify which gamification techniques have been the most helpful in increasing engagement and learning outcomes in cybersecurity education. For example, Abu-Amara et al. (2021) and Weitzl-Harms et al. (2023), display how interactive challenges and real-time feedback can boost user engagement and retention.

II. Assess Educational Results:

The data provides insights into the educational outcomes attained by gamified learning settings. This includes improvements in cybersecurity awareness, practical application of information, and long-term behavioural changes among learners, as emphasised by Beckers and Pape (2016) and Williams et al. (2024).

III. Evaluate Engagements Strategies:

The research evaluates different engagement strategies used in gamified applications, such as leaderboards, achievements or badges, and real-time feedback. These elements are shown to enhance user motivation and foster a competitive yet collaborative learning environment, aligning with findings from Scholefield and Shepherd (2019) and Abu-Amara et al. (2021).

IV. Provide Evidence-Based Recommendations:

The combination of data from diverse sources allowed the researcher to provide evidence-based recommendations for incorporating gamification into traditional methods of cybersecurity education. This involves tailoring learning paths to individual needs and ensuring continual content and updates to guarantee long-term engagement, as recommended by Fatokun et al. (2024) and Murphy and Young (2024).

Through this process, a variety of data types were included, including empirical research, theoretical publications, and industry reports. This broad collection of sources offered a complete view of the current state of gamification in cybersecurity education. The systematic approach to data gathering indicated that the research was supported by strong and diverse evidence, providing with solid foundation for analysis. This method enabled the integration of a diverse set of views and findings, resulting in a more subtle understanding of how gamification can enhance the cybersecurity education and improve social engineering awareness on the overall scale.

3.5 Validity and Reliability

Ensuring the validity and reliability of this research was a critical consideration throughout the study. By using secondary data from reputable sources, this research aimed to provide accurate and trustworthy findings.

3.5.1 Validity:

Validity refers to the accuracy and credibility of the data and the extent to which the research measures what it intends to measure. To ensure validity:

- Source Credibility:

Only peer-reviewed studies, respectable journals, and recognised industry reports were considered. These sources are known for their thorough evaluation processes, which provide legitimacy to the data.

- Relevance:

The literature selection process was driven by precise criteria to guarantee that all sources were directly related to the study goals. This entailed selecting relevant papers published over the previous 10 years that reflect current trends and improvements in cybersecurity education and

gamification.

- Comprehensive Coverage:

A wide range of studies were included by conducting a systematic search using different databases such as Research Gate, Google Scholar, and Microsoft Academic, assuring thorough coverage of the issue.

3.5.2 Reliability:

Reliability refers to the consistency and dependency of the data over time and across different contexts.

To ensure reliability:

- Consistency in Data Collection:

A systematic strategy to data collection was used, which included the use of specified keywords and selection criteria – filters. This guaranteed that the data gathering method was consistent and repeatable.

- Cross-Verification:

Data from several sources was cross-checked to ensure consistency in the findings. For example, comparable themes and conclusions were found in many investigations, increasing the dependability of the findings.

- Detailed Documentation:

The data collecting, organisation, and analysis procedures were well recorded. This extensive documentation enables others to analyse and reproduce the research procedure, boosting the study's credibility.

By adhering to these principles, the research maintains a high standard of validity and reliability. The careful selection of credible and valid sources, systematic data collection methods, and comprehensive coverage ensure that the findings are both accurate and dependable. This strict approach lays correct foundational structure for drawing meaningful conclusions and making informed recommendations based on the research.

3.6 Ethical Considerations

Several ethical factors were carefully considered when performing this research to assure the study's integrity and ethical soundness.

For starters, because the study used secondary data, there was no direct interaction with human participants, hence eliminating the need for informed consent processes. However, it was critical to ensure the ethical application of these secondary data. All data sources were publically available or accessible through academic databases such as Research Gate, Google Scholar, and Microsoft Academic, guaranteeing that the material was collected lawfully and ethically.

3.6.1 Ethical Approval

This research has received an ethical waiver from the Cyber Ethics Panel. As confirmed through email, an emailed dating on 4th of July 2024, ethical approval was waived for this specific study due to the nature of the data being used. The email also reminded the researcher to adhere to the details provided in the complete ethics form and to reapply for approval if any significant changes to the research approach occurred. More information can be found in the appendix section (see Appendix A. - Ethics)

3.6.2 Data Management and Reporting

Given how dependence on secondary data from previously published studies, the initial researchers focused heavily on anonymity and secrecy. This research verified that all references were properly cited while avoiding any revealing of personal information beyond what was publicly accessible. In accordance with ethical reporting standards, all data sources were clearly acknowledged, and suitable citation techniques were used to avoid plagiarism. Each piece of secondary data was thoroughly examined for credibility and relevance, and the conclusions were reported honestly, without any fabrication nor any falsification (Weinbaum et al., 2019; Resnik, D.B., 2020; American Speech-Language-Hearing Association, n.d.).

Chapter 4: Implementation

4.1 Introduction

This chapter presents an in-depth analysis of the data collected from the literature review and the implemented gamified application for cybersecurity education and social engineering awareness. The analysis focuses on evaluating the effectiveness of gamification techniques in enhancing user engagement, motivation, and learning outcomes. By comparing the traditional educational methods with the gamified approach, this chapter aims to highlight the strengths and potential areas for improvement in the current cybersecurity training landscape.

4.2 Implementations of Gamification Techniques

4.2.1 Interactive Challenges

The gamified application's interactive tasks were developed to emulate real-world cybersecurity problems. These challenges stimulate active problem-solving and practical application of information. Abu-Amara et al. (2021) found that interactive learning considerably increases user engagement and long-term retention. According to the findings, individuals who engaged in interactive challenges displayed greater engagement and recall of cybersecurity concepts than those who relied on standard static materials. This study supports Kolb's (1984) experiential learning theory, which emphasises learning via experience.

Key Findings:

- Enhance user engagement and retention.
- Improved practical application of cybersecurity knowledge.
- Positive user feedback on the realism and complexity of scenarios.

4.2.2 Real-Time Feedback

Real-time feedback is another crucial component of the gamified application. By providing instant feedback on user performance, the application helps users understand their mistakes and learn the correct approaches. This method fosters a continuous learning environment where users can quickly adapt and improve their skills. KeepnetLabs Demo and Abu-Amara et al. (2021) highlight the importance of real-time feedback in reinforcing learning and improving performance.

Key Findings:

- Immediate correction of mistakes and reinforcement of correct actions.

- Increased user satisfaction and motivation.
- Higher rates of skill improvement and knowledge retention.

4.2.3 Achievements and Badges

The use of achievements and badges as rewards for completing tasks and reaching milestones serves as a powerful motivator. Inspired by TryHackMe, this technique sustains long-term engagement and interest in the subject matter. Backers and Pape (2016) and Weitzl-Harms et al. (2023) emphasize that rewarding procedures help mitigate competitive stress and enhance user experience.

Key Findings:

- Sustained user engagement over extended periods.
- Increased motivation and sense of accomplishment.
- Positive reinforcement and encouragement of continuous learning.

4.2.4 Leaderboard System

The leaderboard system adds a competitive aspect that pushes users to improve their performance and compare their success to others. This feature promotes a sense of community and collaborative learning, similar to the surroundings established by HackTheBox. Scholefield and Shepherd (2019) and Williams et al. (2024) gave emphasis that leaderboards efficiently address engagement drop-off and maintain user motivation by providing regular updates and new challenges.

Key Findings:

- Increased user motivation through healthy competition.
- Enhanced sense of community and peer learning.
- Reduced engagement drop-off and sustained interest.

4.2.5 Engagement and Motivation

Traditional methods frequently fail to retain user interest and motivation due to their passive nature. In contrast, the gamified approach, which includes interactive challenges, real-time feedback, achievements, and leaderboards, dramatically increases user engagement and motivation. The research supports this conclusion, with various studies such as Abu-Amara et al. (2021) and Beckers and Pape (2016) confirming the efficacy of gamified learning environments.

4.2.6 Practical Application and Skill Improvement

Traditional educational methods often lack practical applications, resulting in lower retention and skill improvement. The gamified approach tackles this issue by providing a hands-on experience in the form of interactive tasks and rapid feedback. This practical application guarantees that users not only acquire theoretical concepts, but also understand how to use them in real-world cases, as demonstrated by Weitz-Harms et al. (2023) and Fatokun et al. (2024).

4.2.7 Inclusivity and Accessibility

The gamified prototype has been developed in a way in order to be inclusive and accessible to both technical and non-technical audiences. Traditional paths most of the times fail to engage non-technical users effectively. Williams et al. (2024) demonstrated that gamification can make cybersecurity education more accessible and engaging for students without a technical background, thereby broadening the reach of cybersecurity training.

4.2.8 Continuous Engagement and Adaptation

Maintaining long-term engagement is a challenge for traditional techniques. The gamified approach, with its regular updates, evolving content, and adaptive learning paths, ensures continuous user engagement. Scholefield and Shepherd (2019) emphasized the importance of regular updates in order to sustain the user's interest, a finding that is reflected in positive feedback from users of the gamified application.

4.3 Conclusion

The findings show that a gamified approach to cybersecurity education and social engineering awareness has considerable advantages over traditional educational techniques. By enhancing user involvement, providing practical learning experiences, ensuring inclusivity, and maintain ongoing engagement, the gamified prototype tackles many of the drawbacks of those traditional techniques. These benefits can be easily provided to the user with the use of main key elements such as interactive challenges, leaderboards, achievements and badges, and real-time feedback. The findings of this analysis, further support the potential of gamification as a revolutionary strategy in cybersecurity education, providing a dynamic and interactive learning environment that better equips individuals to manage the changing landscape of cybersecurity threats.

Chapter 5: Development

5.1 Introduction

This chapter covers the development and implementation of a gamified application designed to enhance cybersecurity education and social engineering awareness. After drawing inspiration from already established industry applications such as HackTheBox, TryHackMe, and KeepnetLabs Demo (Nextdoorsec, 2023), this application was created using Unity and C# programming language, incorporating various gamification elements identified in the literature review (Abu-Amara et al., 2021; Thombre & Velankar, 2022). The primary objective was to create an engaging, interactive learning tool that addresses the shortcomings of traditional educational methods. This application represents the initial prototype of the gamification approach, incorporating the most crucial and significant elements identified in the literature review analysis. Despite the tight timeframe and limited experience, the prototype aims to clearly demonstrate a well-executed gamified strategy by integrating these key components. While not perfect, the focus remains on highlighting the essential features and adapting on the tools available.

5.2 Application Development

The results of the literature review played a significant role on the application development process. Key gamification techniques, such as interactive challenges, real-time feedback, leaderboard, achievements and badges, as well as correct, predefined scoring system were used to increase the user's experience and overall knowledge outcomes. Using elements such as a leaderboard, achievements and badges, which encourage the user to commit more into the overall experience and maintaining further interaction and interest (Thombre & Velankar, 2022). A real-time feedback method with the addition of a well-constructed scoring system that provides the user with feedback preventing the user from getting lost in the process and keeping them intact with the purpose of the application, and finally a variety of interactive challenges engaging with the user and developing more interest between user and education (Beckers & Pape, 2016). Unity's newest UI toolkit was utilized to build a smooth and spontaneous user interface, allowing users to quickly explore and interact with the application. It is highly important to be noted that the prototype will be an executable file in comparison with the ideal setup of a web-based application.

The prototype focused on four key panels that will serve as the Main Scene Panel, Leaderboard Panel, Account Details, Improvement Panel, and lastly Settings Panel. In addition, the prototype will include a traditional Login Panel that the user will be able to use to access their personal account, allowing them to track their progress and maintaining engagement.

5.3 Activity Decision

This gamified approach's activity selection was strategically chosen to address the most important topics in Cybersecurity education: *Physical security*, *Phishing attacks*, and *password complexity*. These topics are fundamental because they target key vulnerabilities that are commonly used as an exploitation for cyberattacks.

Physical Security focuses on educating users about the dangers of tailgating, exposing password that employees tend to misplace, and sharing passwords with others. These simple but often overlooked risks can lead to unauthorized access and significant breaches. Highlighting these aspects can help reinforce the importance of basic but essential physical security measures.

Phishing is essential since it is still one of the most popular and successful social engineering tactics deployed by attackers. Phishing techniques include email spoofing (a technique used in spam and phishing attacks to trick users into thinking a message came from a person or entity they know or trust), malicious links (a link created with the purpose of promoting scams, attacks, and frauds), and impersonation strategies (type of phishing scam where cyber criminals manufacture a sender's email address to make it appear as if the message is from a trusted source) that fool users into disclosing personal information or downloading malicious software. Educating users on recognizing these attacks and responding appropriately is vital in reducing their effectiveness.

Password Complexity is another critical topic since weak passwords are a major source of unauthorised access and breaches. The necessity of choosing strong passwords that include a combination of a minimum length of twelve characters, capital and lowercase letters, numbers, and special characters, as well as avoiding easily guessable passwords, is critical to improving security. Users must also be taught about the dangers of password reuse and the benefits of multi-factor authentication.

By focusing on these topics, the application addresses the fundamental yet critical areas that users must be aware of to maintain a strong cybersecurity posture.

5.4 Login Phase

Starting, upon launching the application, users will encounter a series of standard login procedures designed to streamline account management and access. The application will utilize a database approach to store the account of each user to maintain consistent access. The login phase serves as follows:

I. Login Panel:

Users can enter their personal account by inserting their credentials (username and password), with basic options such as to show the password, retrieve a forgotten password, or create a new account if not already registered.

II. Forgot Password Panel:

This feature allows users to request a password reset link via email, ensuring secure account recovery or in case a user has forgot their credentials. The system will check if the email the user provides matches the initial email the user has provided upon registration to confirm the validity of the request.

III. Create New Account Panel:

Users will be able to provide basic personal information, such as name, surname, email, age range, and nationality, to register a new account and get access into the application.

These initial steps ensure that the application is secure and personalized, setting the stage for a tailored learning experience.

5.5 Main Panels

Once logged in, users can navigate through four main panels using navigation buttons, each serving a specific purpose:

I. Main Scene Panel:

This panel serves as the core learning environment where users can select and complete various levels to earn badges and achievements. Each level is designed to teach specific cybersecurity concepts through interactive challenges. The completion of levels and the accomplished high scores are incentivized to motivate continuous learning (Beckers and Pape (2016); Scholefield and

Shepherd., 2019; Weitzl-Harms et al., 2023). The prototype version will introduce three of the most fundamental aspects of cybersecurity and social engineering, to adapt to the various educational grades of each user. The levels that will be initially introduced to the prototype model:



Figure 2: Main Panel - Preview

a. *Physical Security:*

Introducing users to the fundamentals of physical security, including dangers of tailgating – unauthorized individuals that follow authorized personnel into secure areas (Williams et al., 2024) -, the risks of leaving passwords on plain sight, and the importance of not sharing passwords between other individuals (Abu-Amara et al., 2021).

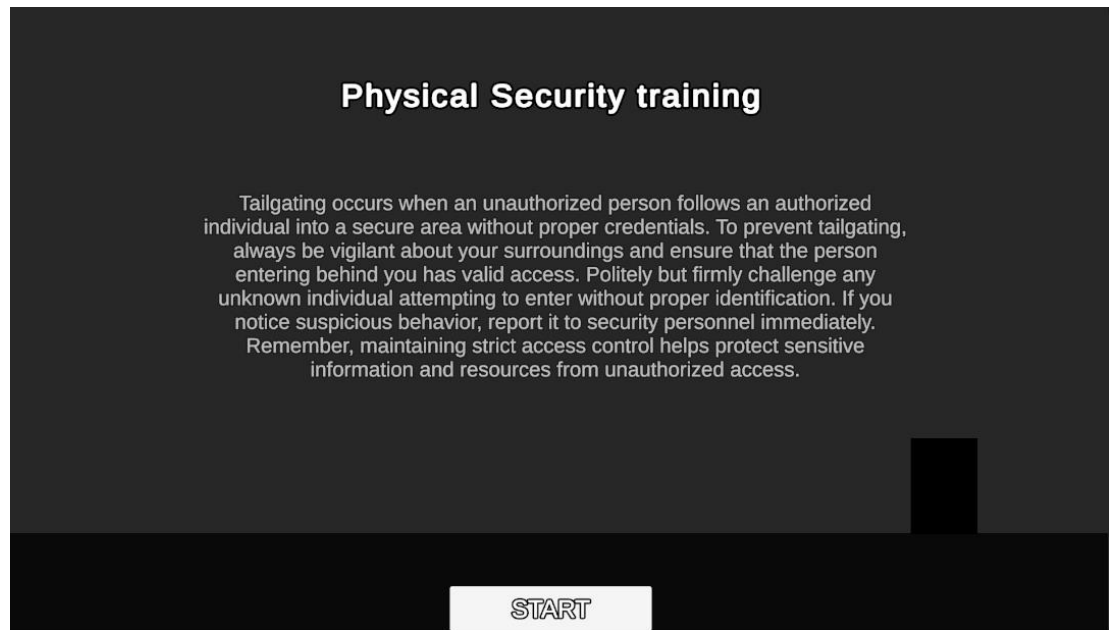


Figure 3: Physical Security - Intro Preview



Figure 4: Physical Security - Activity Preview

b. *Password Complexity:*

A robust tool teaching users regarding the parameters and considerations a proper password should have to avoid password exploitation and brute force attacks.

Empathising the importance of length, complexity – including a mix of letters and capital letters, numbers, and special characters), and avoiding common hazards – such as easily

guessable passwords and password reuse (Hart et al., 2020).

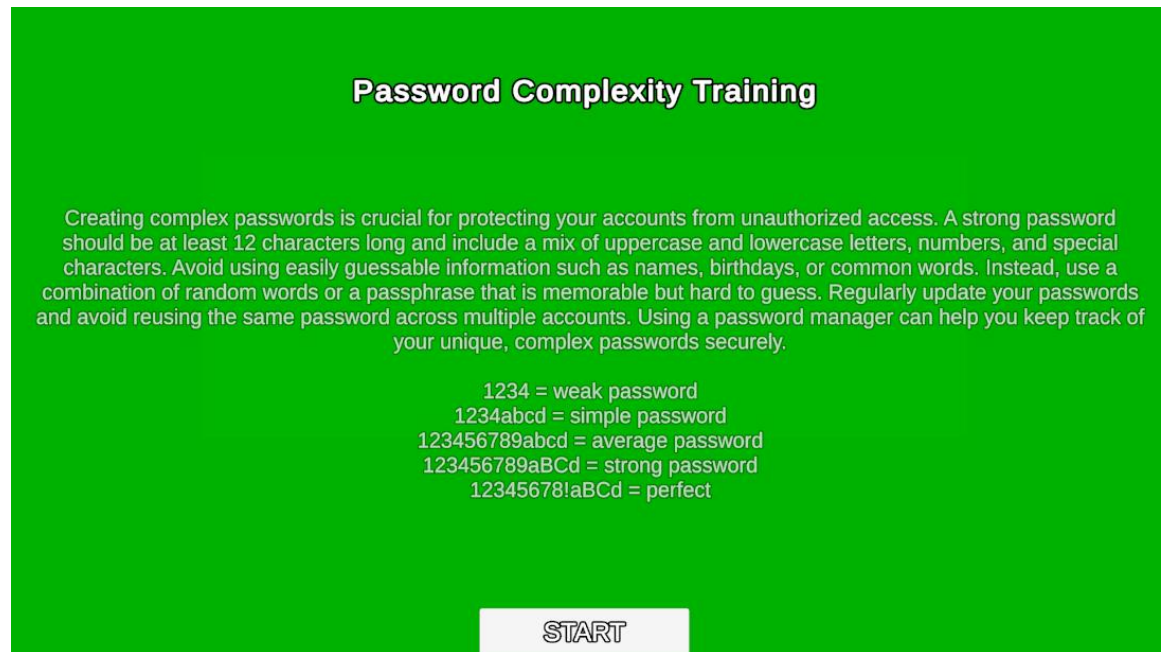


Figure 5: Password Complexity - Intro Preview

Figure 6: Password Complexity - Activity Preview

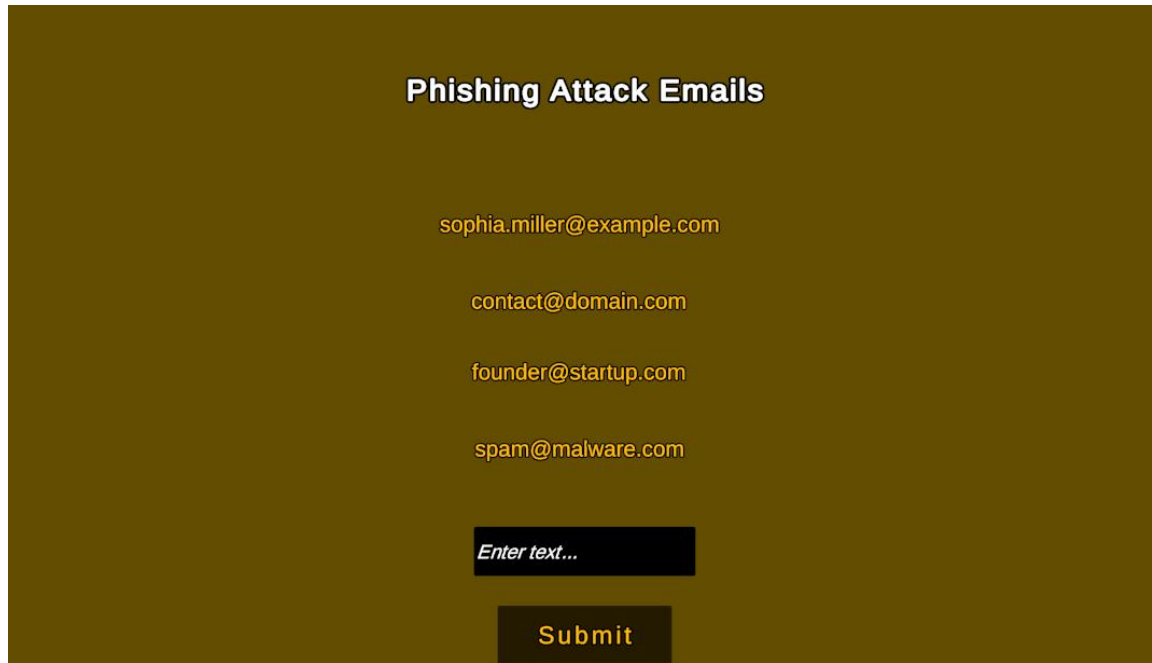
c. *Phishing Attacks:*

As one of the main vulnerabilities – humans – this tool will serve as the guidelines on how a phishing attack can be detected, and how the user should further proceed after receiving a phishing email (Hart et al., 2020) This threat uses psychological methods that trick the user into believing its authenticity, asking for personal details, passwords, or at

some cases installing malicious software that could result in further threats (Murphy & Young, 2024).



Figure 7: Phishing Attack - Intro Preview



d.

Figure 8: Phishing Attack - Activity Preview

II. **Leaderboard Panel:**

This panel introduces a competitive element, displaying the names and ranks of users based on their performance. By comparing their progress with others, users are encouraged to improve their skills and stay engaged with the learning material creating a friendly yet competitive

environment (Nextdoorsec, 2023).

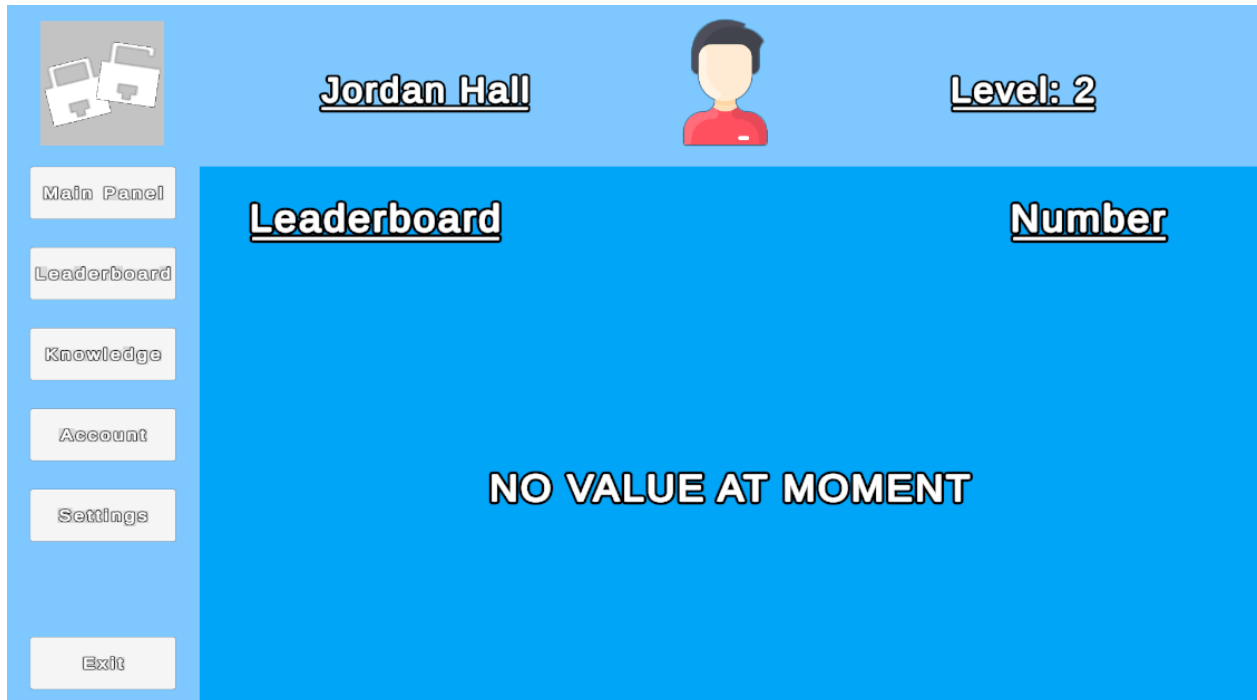


Figure 9: Leaderboard Panel - Preview

III. Account Details Panel:

Users can view their personal information, account details, and the badges and achievements they have earned. User will also be able to edit their personal information from within this panel and change their profile picture to further enhance the aspect of personalization. This panel provides a comprehensive overview of the user's progress and performance, reinforcing their accomplishments and motivating further participation. The importance of allowing the user to personalize the learning experience is supported by Thombre and Velankar (2022), who found that personalized learning paths foster deeper engagement and understanding to the user (Thombre & Velankar, 2022).

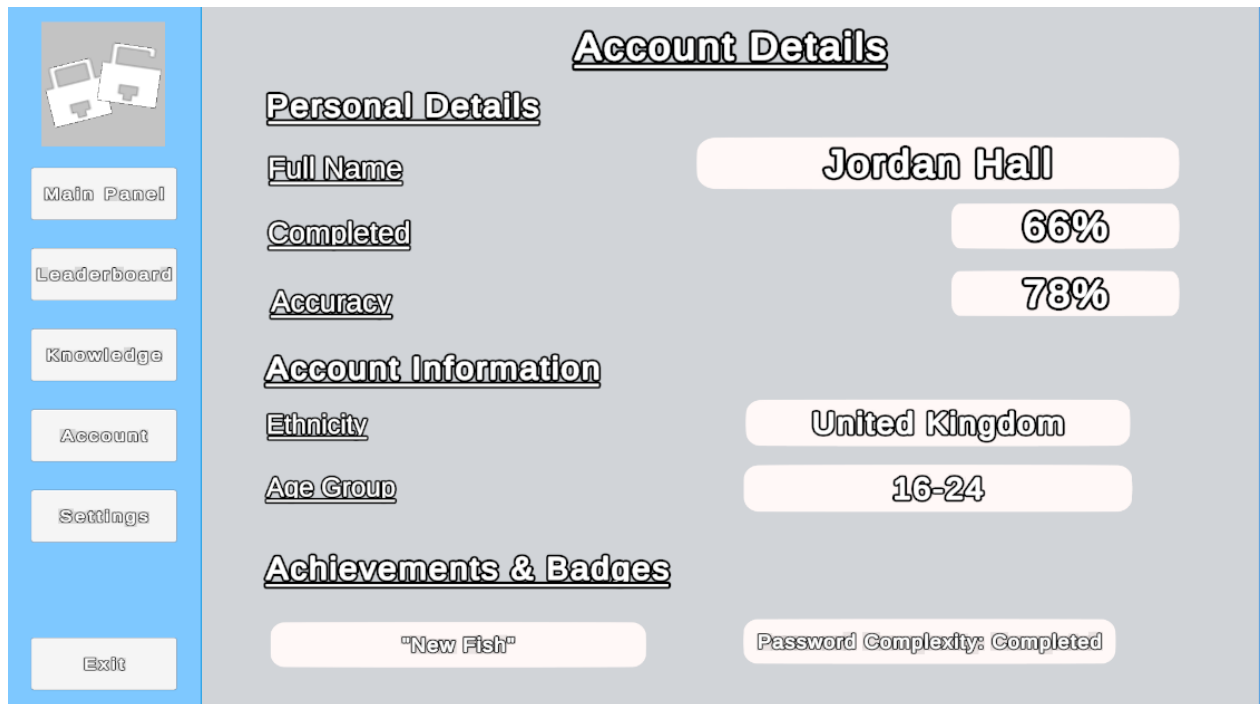


Figure 10: Account Details Panel - Preview

IV. Knowledge Panel:

This highly important panel will implement a crucial component of the gamified approach by providing the users with real-time feedback on their performance for each completed level. This panel – in the prototype stage – will serve as a panel where users will be able to get information and knowledge regarding cybersecurity subject to complete each task. In later versions, upon finishing a level, users will receive insights into their strengths and areas of improvement, a critical aspect for fostering a continuous learning environment. This approach will greatly align with the findings of Abu-Amara et al. (2021) and KeepnetLabs Demo, which emphasizes the importance of real-time feedback in educational gamification for reinforcing learning and improving performance (Abu-Amara et al., 2021; Becker & Pape, 2016).

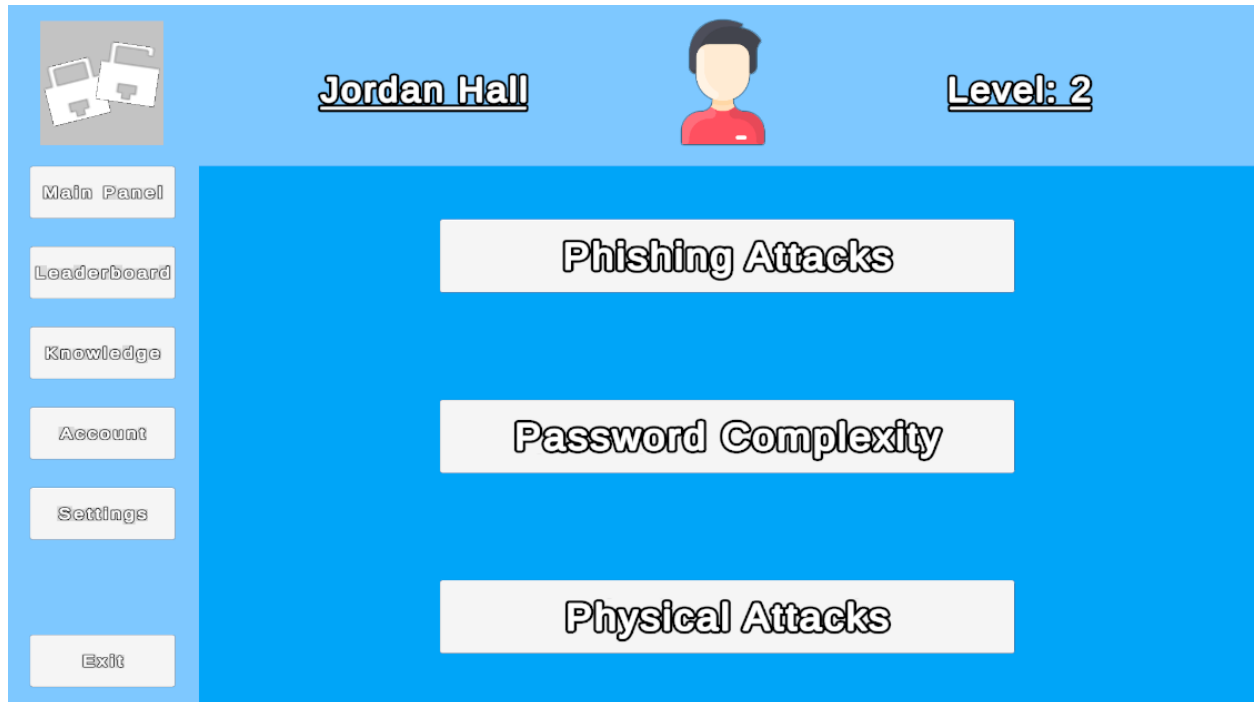


Figure 11: Knowledge Panel (all activities) - Preview

V. Settings Panel:

This panel will simply allow users to customize their experience by adjusting display settings according to their equipment to ensure that it is as user-friendly and as less user-restrictive, sound settings with volume sliders and options to disable click sounds, and other preferences. By offering a modular approach, the application ensures that users can tailor the learning environment to their needs and preferences. The importance of adaptable learning environments is also highlighted by Fatokun et al. (2024), who noted that customizable user preferences help cater to different user scenarios and even further enhance the overall learning experience of the user (Fatokun et al., 2024).

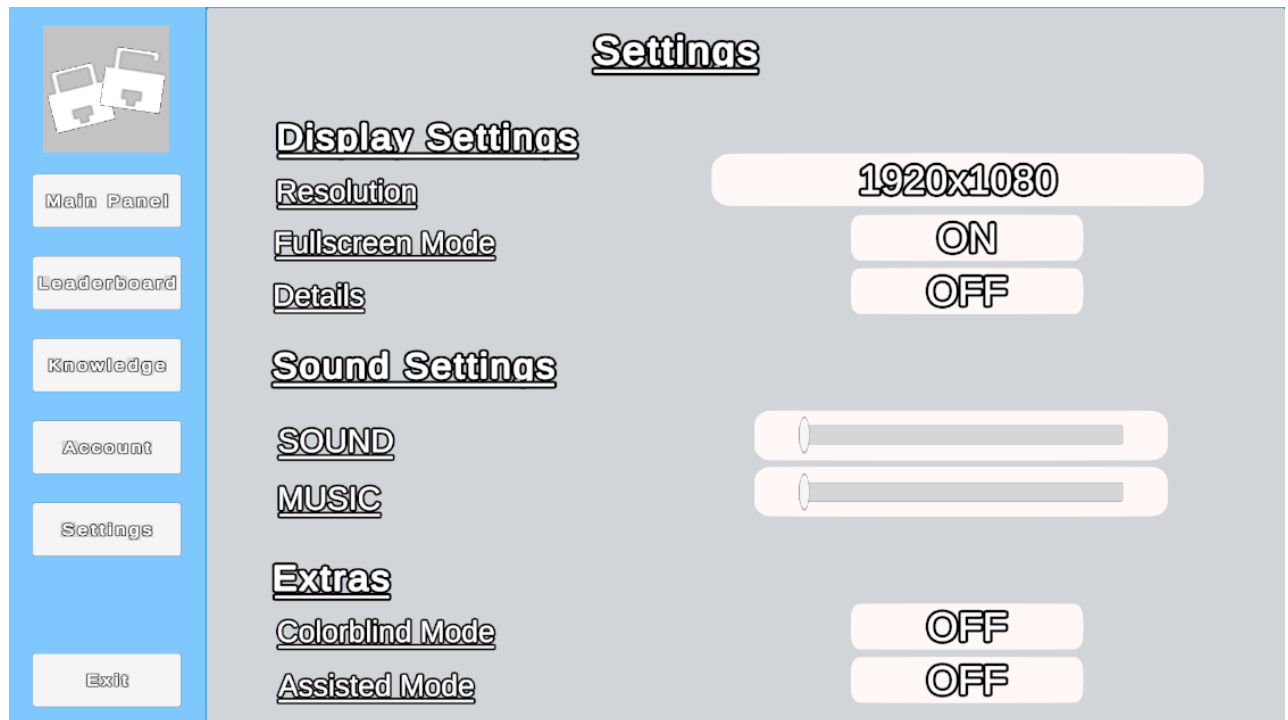


Figure 12: Settings Panel - Preview

5.6 Gamification Techniques

The application utilizes several gamification techniques described in the literature review to improve the learning experience:

- **Interactive Challenges:**

Each level in the 'Main Scene Panel' presents users with real-world cybersecurity scenarios that require active problem-solving, promoting hands-on learning and practical application of knowledge. Abu-Amara et al. (2021) found that interactive learning significantly improves user engagement and long-term retention. This aligns with the findings from Beckers and Pape (2016) and Murphy and Young (2024), who emphasize the importance of realistic and relatable training scenarios to enhance learning outcomes. This will prove to be one of the main implementations in the prototype created to further enhance those learning benefits. (Abu-Amara et al., 2021; Beckers and Pape., 2016; Murphy and Young., 2024)

- **Real-Time Feedback:**

Immediate feedback is provided upon completing challenges, allowing users to learn from their mistakes and understand the correct solutions. This aligns with the approach used by KeepnetLabs Demo, which emphasizes practical engagement and learning from feedback. Additionally, Abu-Amara et al. (2021) and Weigl-Harms et al. (2023) highlights the importance of

clear evaluation metrics, accurate grading measurements, and feedback systems in educational gamification methods (Abu-Amara et al., 2021; Weilt-Harms et al., 2023)

- **Achievements and Badges:**

By earning badges and achievements, users receive recognition for their progress, which serves as a powerful motivator. This technique, inspired by TryHackMe, helps sustain long-term engagement and interest in the subject matter. Beckers and Pape (2016) demonstrated that realistic and engaging educational tools significantly improve user commitment and learning outcomes, while Weilt-Harms et al. (2023) emphasized the role of interactive elements in maintaining user interest. This supported by the findings in the literature review, where balanced competition and rewarding procedures such as badges and achievements help mitigate competitive stress and enhance user experience. (Beckers and Pape., 2016; Weilt-Harms et al., 2023;)

- **Leaderboard System:**

The competitive aspect of the leaderboard encourages users to improve their performance and compare their progress with peers, fostering a community-driven learning environment similar to HackTheBox. Scholefield and Shepherd (2019) noted that leaderboards can effectively address engagement drop-off issues by keeping users motivated through weekly/monthly resets based on addition of regular updates and new challenges, while Weilt-Harms et al., (2023) highlighted the benefits of competitive elements in sustaining user engagement (Scholefield and Shepherd., 2019; Weilt-Harms et al., 2023). Additionally, the leaderboard helps manage initial paranoia by weekly or even monthly resets, encouraging continuous improvement while promoting healthy competition, ensuring sustain interest without overwhelming users, as suggested by Murphy and Young (2024) and Breta et al. (2017). (Murphy and Young., 2024; Breta et al., 2017)

Chapter 6: Evaluation of Implementations

This section evaluates the practical implementations of the gamified application designed to address the traditional shortcomings of cybersecurity education. By implementing interactive scenarios, immediate feedback, achievements and badges, adaptive learning paths and community-driven competition, the application aims to enhance user engagement, motivation, and retention of cybersecurity knowledge.

6.1 Introduction

The traditional methods of cybersecurity education, often reliant on static materials such as articles, videos, and lectures, have proven insufficient in effectively engaging users and fostering long-term behavioural changes. This thesis explores the potential of gamification as an innovative approach to address these shortcomings. The literature reviewed encompasses various studies and applications of assesses whether the implemented gamification techniques achieve their intended goals of enhancing engagement, providing practical learning experiences, and ensuring continuous and inclusive education.

6.2 Enhanced Engagement and Motivation

Implementation: Interactive Scenarios and Rewards

- **Interactive Scenarios:**

Users are able to engage in real-world cybersecurity challenges that simulate potential dangers, providing with a more hands-on experience and improving problem-solving skills. Abu-Amara et al. (2021) demonstrated that interactive learning significantly enhances users' engagement and long-term knowledge retention. This implementation ensures that learners are not just passively consuming information but actively applying it.

- **Achievements, Badges and Progress Tracking:**

The prototype utilizes achievements, badges and progress tracking systems that reward users upon the completion of a task and reaching milestones. This recognition helps maintaining motivation and provides a sense of accomplishment, encouraging continuous engagement. Beckers and Pape (2016) noted that engaging educational tools significantly improve user commitment and learning outcomes. One of the reasons of the high importance in the utilization in the prototype.

6.3 Active Learning and Practical Application

Implementation: Real-Time Feedback Mechanism

- **Real-Time Feedback:**

By providing instant feedback, the prototype aids users understand their mistakes and learn the correct approaches. This strategy creates a learning environment which users may consistently enhance their abilities through real-time assessments. KeepnetLabs Demo illustrated the importance of prioritising practical involvement and learning from feedback. Something that is also supported by other literature review, which aligns with the main objective of reinforcing learning and improving performance. (Fatokun et al., 2024)

6.4 Customized Learning Paths and Feedback

Implementation: Adaptive Learning Paths

- **Adaptive Learning Paths:**

The prototype allows the user to select various challenges depending on their knowledge and skills in hand. Providing with a more tailored approach that allows the user to enjoy the gamification technique, while ensuring that the learning experience is neither too easy nor impossible. This ‘flexibility’ accommodates different learning speeds and demands. As previous examples, such as Fatokun et al. (2024) used the Technology Thread Avoidance Theory (TTAT) to create a complete gamified learning tool that, ensuring the content remains relevant and effective against evolving cyber threats. While not implemented in the prototype, benefits from the Technology Thread Avoidance Theory (TTAT) have been taken into consideration in the implementation part.

6.5 Inclusivity and Accessibility

Implementation: Inclusive Design and Content

- **Inclusivity and Accessibility:**

Williams et al. (2024) extended the discussion to non-technical audiences, proving that gamification can make cybersecurity education accessible and engaging for students and employees without a technical background. The prototype is designed to be inclusive, ensuring that foundation cybersecurity principles are understood by a wide audience, making cybersecurity education more accessible and engaging for all learners.

6.6 Continuous Engagement and Content Updates

Implementation: Community and Competition

- **Community and Competition:**

The leaderboard system – which serves as one of the key implementations – introduces a competitive element, encouraging users to improve their skills to rank higher. This feature fosters a sense of community and share learning, similar to other community-driven environments such as HackTheBox and TryHackMe. In addition, Scholefield and Shepherd (2019) suggested that regular updates and evolving content to sustain user interest, is pivotal. New material can result in new high scores through the leaderboard, resulting in a stronger bond between the users of the community. Encouraging collaboration and knowledge sharing, which is crucial for a comprehensive learning experience. (Murphy & Young, 2024)

6.7 Summary

The evaluation of these various implementations display that the gamified application can effectively address the traditional shortcoming of the pass years regarding cybersecurity and social awareness education. By increasing engagement, providing more practical learning experiences, ensuring adaptive and inclusive education, and encouraging ongoing participation, the prototype meets its intended goals. These findings underscore the potential of gamification as a revolutionary method in cybersecurity education, offering a dynamic and interactive learning environment that better prepares individuals to navigate the evolving landscape of cybersecurity threats.

Chapter 7: Discussion and Conclusion

7.1 [Discussion](#)

7.1.1 Interpretation of Findings

A major step forward in addressing the shortcomings of conventional teaching approaches is the creation of a gamified cybersecurity education prototype. The results show that gamification significantly improves user motivation, engagement, and retention of knowledge. By implementing interactive challenges, achievements/badges, leaderboards, and real-time feedback, traditional methods become less static and promote a more dynamic learning environment. This is in line with the studies by Beckers and Pape (2016) and Abu-Amara et al. (2021) that showed how successful gamification can be in educational settings. By ensuring that both technical and non-technical users may benefit, the inclusive design expands the audience for cybersecurity education.

7.1.2 Limitations of the study

There are several limitations to this study that affect how deep and broadly applicable the conclusions are. Reliance on secondary data restricts the analysis to existing information, limiting the direct observations of user engagement and behaviours. Moreover, this research' participant samples were frequently consistent, which limited the results' generalisability across a range of demographics. The technological constraints of the prototype, developed with basic tools like Unity and C#, also limited the integration of advanced features that could enhance the learning experience. Finally, the limited evaluation period only permitted the assessment of knowledge retention and short-term engagement, leaving open issues regarding long-term efficacy. Future research must address these issues in order to get a more thorough knowledge of gamified cybersecurity education.

7.1.3 Suggested Improvements for Future Implementations

This section outlines key areas for enhancing the gamified approach, including UI/UX design, more diverse challenges, and personalized learning paths. These improvements can optimize user engagement and learning effectiveness in the cybersecurity education application:

- I. **More Detailed Application Design:** Future versions should focus on a more sophisticated and visually appealing interface, employing advanced UI/UX principles to enhance user satisfaction and learning outcomes (Weitl-Harms et al., 2023).
- II. **Enhanced Colour Palette:** Utilize a vibrant and strategically chosen colour scheme to maintain high levels of user motivation and engagement (Nextdoorsec, 2023).

- III. **Expanded Gameplay Features:** Introduce a wider range of challenges, including time-based, multiplayer, and cooperative activities, to cater to a broader audience and increase engagement (Abu-Amara et al., 2021).
- IV. **Incorporation of Real-World Scenarios:** Include more detailed and up-to-date real-world scenarios to make the learning experience more practical and relevant to the user (Becker and Pape, 2016).
- V. **Advanced Feedback Mechanisms:** Enhance the feedback mechanism to provide user-specific, adaptive feedback that takes into account their learning preferences and progress (KeepnetLabs Demo)
- VI. **Comprehensive Knowledge Base:** Include a comprehensive knowledge base that offers in-depth information on a range of cybersecurity subjects within the application (Weitl-Harms et al., 2023; Abu-Amara et al., 2021).
- VII. **Regular Content Updates:** By frequently adding new challenges, scenarios and more educational resources to the content, ensuring that the application remains interesting and relevant (Scholefield & Shepherd, 2019; Williams et al., 2024).
- VIII. **Community Building Features:** Foster a sense of community through forums, chat rooms, blogs, and collaborative challenges, enhancing the overall learning experience (Nextdoorsec, 2023)
- IX. **Enhanced Personalization:** Employing sophisticated personalisation tools to customise the learning process to each user's unique requirements, guaranteeing effective and enjoyable learning outcomes (Thombre & Velankar, 2022).
- X. **Accessibility Improvements:** Making the application even more accessible to users with disabilities by incorporating features such as screen readers, scalable text, and alternative input methods (Williams et al., 2024; Fatokun et al., 2024).

7.1.4 Future Research Directions

Several areas warrant further research to build on the findings of this study:

- I. **Longitudinal Studies:** Conduct long-term studies to assess the sustainability of engagement and knowledge retention over extended periods.
- II. **Diverse Participant Pools:** Expand the diversity of study participants to include various demographic groups, ensuring that the findings are generalizable across different populations.
- III. **Technological Advancements:** Examine how to apply cutting-edge technology like artificial intelligence (AI) and virtual reality (VR) to improve gamified learning even further.

- IV. Behavioural Impact: Investigate the long-term behavioural impact of gamified cybersecurity education on users' real-world cybersecurity practices.
- V. Comparative Studies: To determine which gamification solutions are best for cybersecurity education, compare and contrast various approaches.
- VI. User Feedback Integration: Keep the development process sensitive to user demands and preferences by incorporating user feedback on a regular basis.

In order to guarantee that gamified cybersecurity education stays a useful and efficient tool in the fight against cybersecurity risks, future research may continue to hone and broaden its potential by tackling these topics.

7.2 [Conclusion](#)

Gamification has the ability to completely revolutionise traditional educational methods, as evidenced by the research and the development of the gamified cybersecurity education application. The results demonstrate that gamification may greatly increase user engagement, motivation, and knowledge retention. Users receive an engaging and effective hands-on learning experience when theoretical principles are practically applied through interactive scenarios. Furthermore, a wider audience may benefit from cybersecurity education due to the gamified approach's accessibility and inclusion, which promotes a culture more conscious of cybersecurity threats.

However, to keep the application relevant and effective, ongoing adaptation and more enhancement are essential. Future improvements can further improve the overall learning experience of the users. These could include more intricate application design, a better colour palette, expanded gameplay features, real-world scenarios and challenges incorporation, sophisticated feedback mechanisms, extensive knowledge bases, frequent content updates, enhanced personalisation, community building features, and accessibility improvements.

Ultimately, the gamified approach to cybersecurity education presents a dynamic and innovative solution to the evolving challenges of cybersecurity threats. By fostering an engaging and interactive learning environment, it equips individuals with the knowledge and skills needed to navigate the complex landscape of cybersecurity, contributing to a safer digital world.

Chapter 8: Reference list

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Appendix A: Material/Details that may be needed

Ethics - Prove of Knowledge:

Necessary material in relation to ethical approval and ethical certifications:



Figure 13: 23-24 Research Ethics Badge



Figure 14: 2024 Ethics in Research Badge

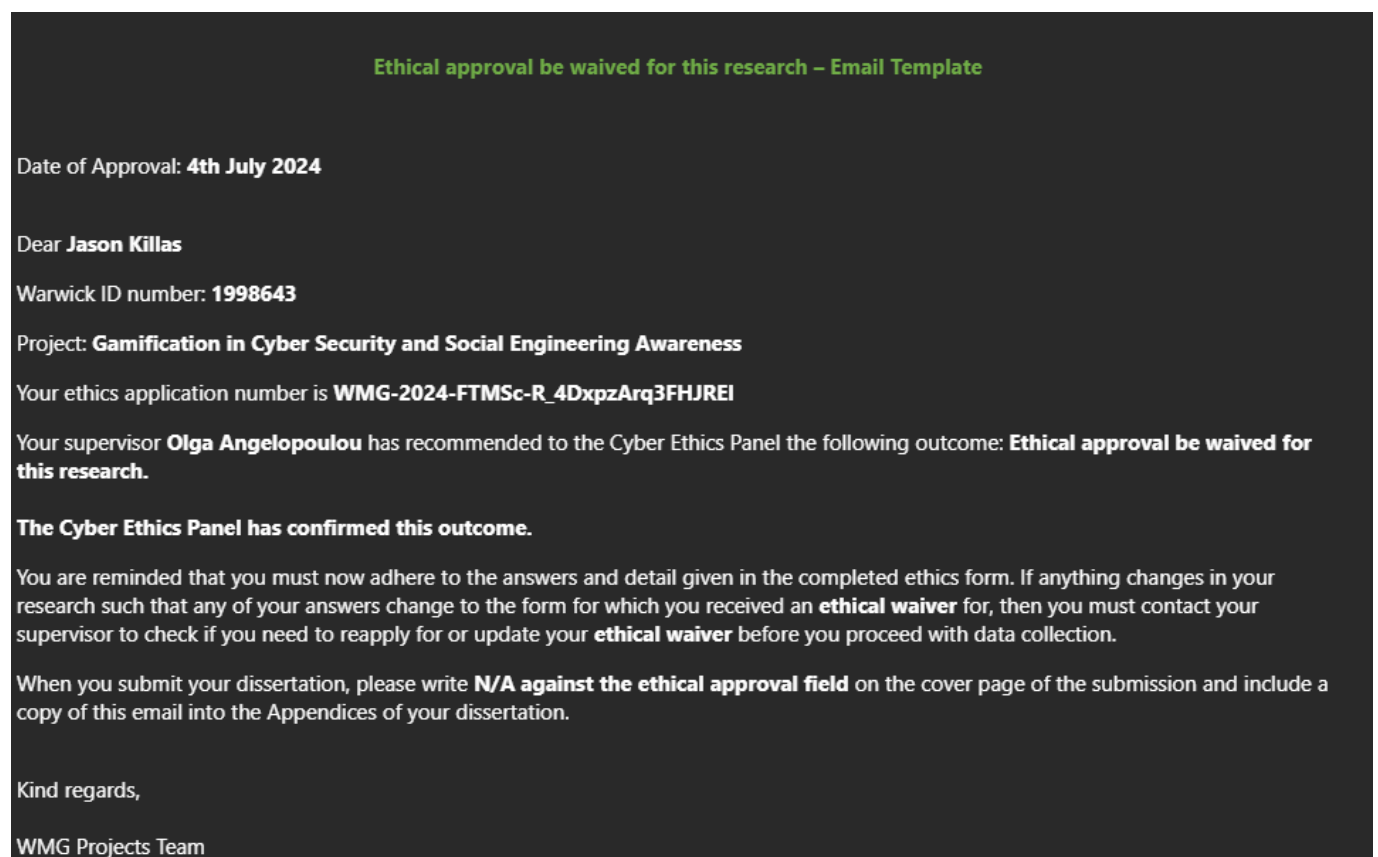


Figure 15: Ethical Approval Email - Waiver

Technology Threat Avoidance Theory (TTAT):

Technology Threat Avoidance Theory (TTAT) is a conceptual framework that examines how people react to and avoid technology threats, especially around cybersecurity. Developed by Liang and Xue (2010), the key concepts include perceiving vulnerability, perceived severity, perceived danger, effectiveness of preventive, cost of prevention and self-efficacy from the foundation of TTAT. These concepts affect a person's desire to participate in avoidance actions aimed at reducing or eliminating cybersecurity risks. According to the hypothesis, perceived severity (the gravity of a threat's repercussions) and perceived susceptibility (the likelihood of encountering a threat) are important indicators of total perceived danger, which in turn greatly encourages people to take preventive actions. The study revealed that beliefs regarding the effectiveness of preventive measures, their associated costs, and confidence in executing them do not significantly influence motivation to avoid cybersecurity threats in the context of information sharing. This is interesting because the study found that self-efficacy, prevention effectiveness, and prevention cost were not significant predictors of avoidance motivation. Organisations and policymakers can gain valuable insights from the application of TTAT in cybersecurity. It suggests that raising perceptions of susceptibility and severity could motivate professionals to share important cybersecurity information and work together on preventive measures, improving overall strategies for mitigating threats.

GitHub Repository Link:

<https://github.com/JasonK00/MSc-Cyber-Security-Engineering>

Additional Case Studies:

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